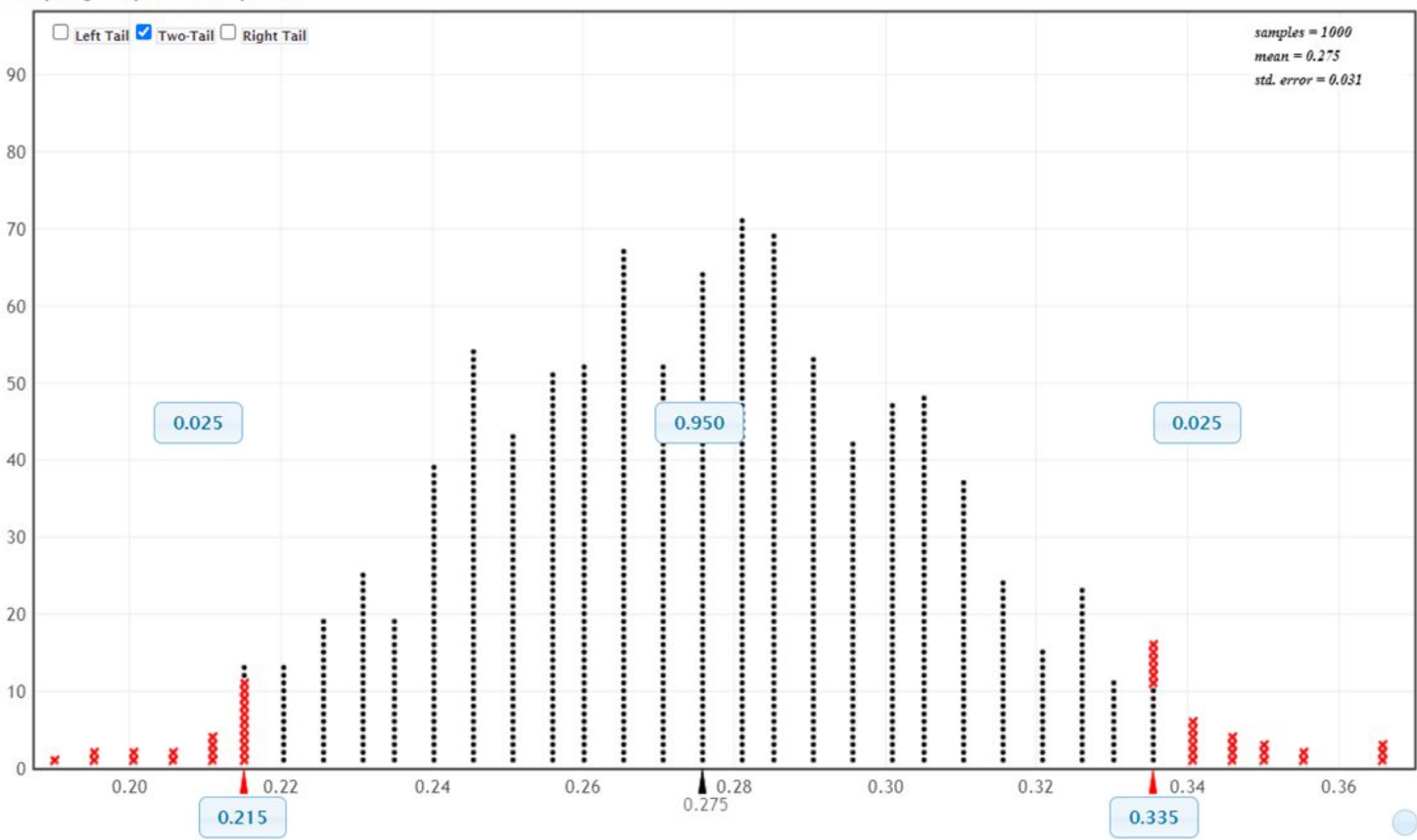


Idea of confidence interval, continued

MATH 213

Demo: How to construct an interval of plausible values for the population parameter.

Sampling Dotplot of Proportion



Q1: What percentage of the sample proportions in this dotplot will have intervals (constructed as demonstrated above) that DO CONTAIN the population proportion of 0.275?

95% of them (the black dots) will have confidence intervals that contain the population proportion.

(Note: little bit of ambiguity with the proportions that have both black and red dots in it, but this is a small number)

Q2: What percentage of sample proportions in this dotplot will have intervals (constructed as demonstrated above) that DO NOT CONTAIN the population proportion of 0.275?

5% will not!

BIG PROBLEM: In practice, we have no idea what the sampling distribution looks like. We can't afford to take thousands and thousands of samples to get a picture of the sampling distribution. (If we did, we'd know the population proportion because it would be at the center of that distribution.) So we don't know what the percentiles of the sampling distribution are, so we can't construct an interval.

How do we get around this? A few different ways. One way, for today: **bootstrapping**.

Bootstrapping - The Idea:

- Select one sample of size 200 using random sampling.
- Imagine having many, many copies of that sample. As long as the original sample was selected randomly, we would expect this to estimate the population.
- Then sample repeatedly from our estimated population to make an approximate sampling distribution called the bootstrap distribution, find the percentiles, compute a 95% confidence interval. There are theorems from mathematics that say the bootstrap distribution has similar properties to the sampling distribution so using the percentiles from the bootstrap distribution is a good idea.

Bootstrapping - In Practice:

- Making multiple copies of the sample is difficult.
- After selecting one sample, instead of making multiple copies of it, we sample from it *with replacement* until we have selected as many subjects as were in the original sample. This is called a bootstrap sample. To make a bootstrap sampling distribution, compute the bootstrap proportion and plot it on the dotplot. Do this over and over again until we have a nice full dotplot called the bootstrap distribution. Compute the percentiles and make the confidence interval.

